

Fault Positioning Method for Distribution Systems Built upon Positive Sequence Fault Components

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Abstract—This paper addresses the challenge of fault location for interphase faults in medium and low voltage distribution networks by innovatively integrating the positive sequence fault component into the double-ended impedance method, thereby proposing a novel fault location technique based on this component. In the event of a fault in a distribution network system powered by dual sources, this method isolates the positive sequence components of the fault quantities of bus voltages and currents from both sides. Thereafter, the dichotomy method is utilized to hypothesize the fault point. Utilizing the line parameter equation and the positive sequence fault components of the voltages and currents on both sides, the positive sequence fault component voltages at the presumed fault location are calculated individually. By comparing the two computed positive sequence fault component voltages, the range is progressively refined to converge on the actual fault location, thus yielding a relatively precise fault distance. The fault location method presented in this paper exhibits a commendable efficacy in practical applications, with the fault location error not surpassing 2% of the total line length.

Keywords—medium and low voltage distribution network, interphase fault, positive sequence fault component, fault location, dichotomy method

I. INTRODUCTION

The power system's distribution network is a critical component, its secure and stable operation holds great importance in guaranteeing power supply reliability and enhancing power quality. Among various factors, faults are the primary ones that affect the reliable operation of the distribution network [1]. Neutral point ungrounded mode or arc suppression coil neutral point grounding is the method most commonly used in China's medium and low voltage distribution networks. Due to its specific features, a single-phase grounding fault has a negligible effect on the power grid's overall operation. In contrast, interphase faults are common and thorny issues. Interphase faults may lead to line outages and equipment

damage, thus affecting the normal power consumption of a large number of users.

Precise fault location can assist operation and maintenance personnel in quickly pinpointing the fault points, reducing the time spent on fault detection, and improving the efficiency of fault repair. As a result, it can effectively mitigate the duration of power outages and economic losses, ensure the distribution network's dependable power supply, and play a critical role in the preservation of the normal flow of social production and daily life. [2-3].

Distribution network lines are comparatively short, demanding high precision in fault location. Additionally, due to its intricate and varied structure along with a multitude of influencing factors, the complexity of fault location in the distribution network is further exacerbated [4]. Currently, the fault-location techniques for distribution networks both domestically and internationally can be categorized, based on the measurement ends, into the single-end method, double-end method, and multi-end method. That is, they utilize the information from one end, two ends, or multiple ends of the line to conduct fault location. In terms of principles, they can be roughly classified into the impedance-based method, traveling-wave method, S-injection method, and other intelligent fault-location algorithms [5].

Implementing the impedance method [6-7] is effortless due to its simplified principle. Nevertheless, it is influenced by variables such as transition resistance and load current. The traveling wave method [8-10] features a fast fault location speed and high accuracy, and it is not influenced by factors like transition resistance and the operating mode of the power system. Nevertheless, it has high requirements for hardware, necessitating the installation of specialized traveling wave detection equipment. Moreover, the traveling wave will experience attenuation and distortion during the propagation process, that could influence how precise the fault locator is. To

pinpoint the exact location of a fault using the injection method, a signal detector follows the path that a specially designed signal takes as it travels through the line. The process of injection [11-12] incurs high costs, and the injected signal may be disrupted by factors like line loss and noise. Other intelligent fault-location algorithms [13-15] pertain to location methods that make use of modern computer technology and data-analysis tools. These algorithms demand a vast amount of training data and computing resources.

Positive-sequence fault components are not constrained by fault types. They remain unaffected by load current, transition resistance, and system electromotive force. These components are capable of nullifying the effect of the mutual inductance that exists between parallel lines upon fault location. Moreover, the processes of extracting and analyzing positive sequence fault components are comparatively straightforward.

This paper addresses the challenge of fault location for interphase faults in medium and low voltage distribution networks by innovatively integrating the positive sequence fault component into the double-ended impedance method, thereby presenting a method for locating defects that utilizes this element. This technique separates the positive sequence components of the fault quantities of the bus voltages and currents from both sides in the case of a fault in a distribution network system that is powered by dual sources. The next step is to use the dichotomy method to guess where the problem is. The individual voltages of the positive sequence fault components at the assumed fault point are determined using the line parameter equation in conjunction with the currents and voltages on both sides. A fairly accurate fault distance is obtained by gradually refining the range until it converges on the real fault location, which is achieved by comparing the two computed positive sequence fault component voltages. This paper presents a fault location method that works very well in real-world scenarios, with the fault location error remaining below 2% of the total line length.

II. BASIC PRINCIPLES OF THE DISTANCE MEASUREMENT ALGORITHM

A. Model Establishment and Data Selection

As the lines in the distribution network are relatively short and the distributed parameter characteristics are not prominent, to simplify the calculation, this paper employs the π -type line equivalent model. Fault components arise solely after a fault transpires, with their value corresponding to the disparity between the electrical quantity post-fault and that pre-fault. This attribute guarantees that the fault component remains unaffected by load current, transition resistance, or system voltage. The zero-sequence and negative-sequence fault components are frequently constrained by the fault type. Nonetheless, the positive-sequence fault component is markedly dissimilar. Irrespective of the fault type that manifests on the line, it is invariably present and not constrained by the nature of the fault. This paper opts to employ the positive-sequence fault component for fault location tasks based on this advantage.

Fig. 1 is the circuit diagram of a medium and low voltage distribution network supplied by a two-terminal power source. Under normal operating conditions, the voltage at any point

along line MN can be computed separately using the electrical parameters from the M and N terminals. For the same location on the line, the two calculated voltages are identical. Similarly, if a fault were to happen at a specific point on the line, the positive sequence fault voltage computed using electrical quantities at the M terminal and the N terminal would be identical.

Assume the line is homogeneous. Denote r_1 , l_1 , and c_1 as the positive-sequence resistance, inductance, and capacitance per unit length of the line respectively.

Subsequently, the positive sequence fault component voltages at the fault point calculated from both the M terminal and the N terminal, are presented in (1) and (2) respectively.

$$\Delta \dot{U}_{M1.F} = \Delta \dot{U}_{M1} - \left(\Delta \dot{I}_{M1} - \frac{\Delta \dot{U}_{M1}}{j\omega c_1} \frac{L_F}{2} \right) \cdot (r_1 + j\omega l_1) \cdot L_F \quad (1)$$

In (1), $\Delta \dot{U}_{M1}$ and $\Delta \dot{I}_{M1}$ represent the positive-sequence components of the voltage and current fault quantities on the M side respectively. L stands for the total length of the line, and L_F stands for the distance from M-terminal to the fault point.

$$\Delta \dot{U}_{N1.F} = \Delta \dot{U}_{N1} - \left(\Delta \dot{I}_{N1} - \frac{\Delta \dot{U}_{N1}}{j\omega c_1} \frac{L - L_F}{2} \right) \cdot (r_1 + j\omega l_1) \cdot (L - L_F) \quad (2)$$

In (2), $\Delta \dot{U}_{N1}$ and $\Delta \dot{I}_{N1}$ represent the positive-sequence components of the voltage and current fault quantities on the N side respectively.

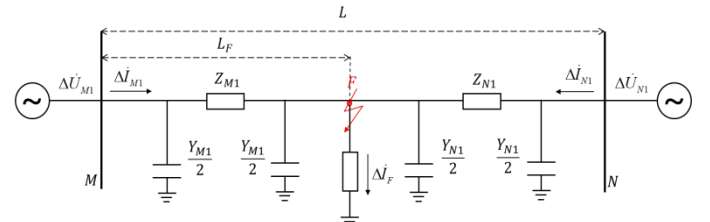


Fig. 1. Circuit diagram of medium and low voltage distribution network powered by double-ended power supplies

B. Specific Implementation Method of Distance Measurement

When the fault distance L_F is undetermined, as depicted in Fig. 2, the bisection method is utilized to identify the fault point.

The binary method, also known as binary search or half-interval search, is a classic algorithm that can efficiently locate a specific element within an ordered array. Its core operating mechanism involves continuously halving the search scope to gradually narrow down the search interval until the target element is successfully found or it is determined that the target element does not exist. During each iteration, the algorithm precisely calculates the index of the middle element in the array. Subsequently, it compares this middle element with the target element. Based on the comparison result, the algorithm can "intelligently" decide whether to continue the next round of

search in the left or right half of the array. With its low time-complexity, binary search can complete the search task rapidly and accurately, making it one of the preferred algorithms for solving numerous data search problems.

Assume that the fault occurs at the midpoint of the line (point L_1 in Fig. 2). Employ the positive-sequence fault components of the voltages and currents on both sides of the bus to determine the two positive-sequence fault-component voltages at the designated fault location.

Next, separately compare the phase angles and amplitudes of these two voltages, and calculate the amplitude difference and phase-angle difference. Assess whether the fault-location result satisfies the established criteria by examining the phase-angle difference and amplitude difference. The assumed fault point's position at this stage is the actual fault location if the result meets the requirements. Conversely, if the requirements are not met, determine the direction for the subsequent search of the fault point according to the leading or lagging relationship of the phases.

When the fault distance L_F is undetermined, the bisection method can be employed to locate the fault point, as illustrated in Figure 2. Initially, assume that the fault point is at the mid-point of the line ($L_F = 1/2 L = L_1$). Two voltages representing positive-sequence fault components are determined for the assumed fault point based on the current and voltage components on either side of the bus. The difference in phase angle and amplitude are then calculated by comparing the two voltages' phase angles and amplitudes. When the phase angle difference and amplitude difference are found to be within the acceptable range for fault location, the process proceeds to the next step. In that case, the presumed location of the point of failure is ideal. In such case, the leading or lagging relationship between the phases of the two voltages dictates the next search direction for the point of failure.

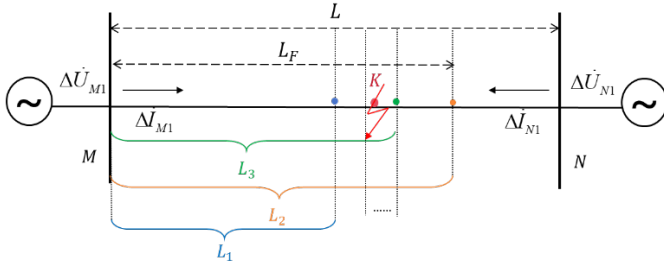


Fig. 2. The process diagram of using the bisection method to find the fault point

$$\begin{cases} \Delta\varphi = \Delta\varphi_{M1.F} - \Delta\varphi_{N1.F} \\ \Delta U = \Delta U_{M1.F} - \Delta U_{N1.F} \end{cases} \quad (3)$$

When condition (4) holds true, the two voltages are regarded as equal. In such a case, the hypothesized fault point is deemed to be the actual one.

$$\begin{cases} |\Delta\varphi| \leq 5\% \frac{|\Delta\varphi_{M1.F}| + |\Delta\varphi_{N1.F}|}{2} \\ |\Delta U| \leq 5\% \frac{|\Delta U_{M1.F}| + |\Delta U_{N1.F}|}{2} \end{cases} \quad (4)$$

In (4), $\Delta\varphi_{M1.F}$, $\Delta U_{M1.F}$, $\Delta\varphi_{N1.F}$ and $\Delta U_{N1.F}$ stand for the phase angle and amplitude of the positive sequence fault component voltage at the assumed fault point derived from both sides of the bus respectively.

If the convergence condition is not met, it is necessary to determine the next search direction for the fault point based on the leading or lagging relationship of the phases. When $\Delta\varphi > 0$, it means that the positive sequence fault component voltage at L_1 derived from the M side leads in phase to that derived from the N side. At this time, the actual fault point K is located at a position closer to the N side from the mid-point of the line. Next, the bisection method is used. The mid-point of the right-hand half of the line is selected as the new assumed fault point. The two positive-sequence fault component voltages corresponding to this point are recalculated, and their phase angles and amplitudes are compared again.

Conversely, when $\Delta\varphi < 0$, it indicates that the positive-sequence fault component voltage at L_1 derived from the M side lags in phase behind that derived from the N side. In this case, the actual fault point K is located at a position closer to the M side from the mid-point of the line. Similarly, the bisection method is applied. The mid-point of the left-hand half of the line is chosen as the new assumed fault point. The two positive-sequence fault component voltages at this point are recalculated, and then the comparison of the phase angles and amplitudes is carried out again.

By continuously iterating using the aforementioned method, the search scope is gradually narrowed, thereby steadily approaching the actual location of the fault point.

III. THE SPECIFIC PROCESS OF FAULT LOCATION FOR DISTANCE MEASUREMENT

The flowchart of the fault location algorithm proposed in this paper is shown in Figure 3, and it specifically includes the following steps:

- Collect the three-phase voltages and three-phase currents at the M side before and after the fault, as well as the three-phase voltages and three-phase currents at the N side.
- The Kalman filter algorithm is adopted. By continuously updating the estimated value and the error covariance matrix, the noise and harmonics in the signal are effectively removed. After that, it is represented in the form of phasors through the Fourier transform.
- Employ the symmetrical component approach to isolate the positive sequence components $\dot{U}_{M1}$, $\dot{U}_{N1}$, $\dot{I}_{M1}$, $\dot{I}_{N1}$, $\dot{U}'_{M1}$, $\dot{U}'_{N1}$, $\dot{I}'_{M1}$, $\dot{I}'_{N1}$ of the voltages and currents on both sides prior to and subsequent to the fault.

- Deduct the positive sequence voltages and currents in the pre-fault condition from their counterparts in the post-fault condition. This enables the acquisition of the positive sequence fault components $\Delta \dot{U}_{M1}$, $\Delta \dot{U}_{N1}$, $\Delta \dot{I}_{M1}$, $\Delta \dot{I}_{N1}$ of the bus voltages on both sides.
- The bisection method is used to search for the fault distance. Assume that the fault distance is L_k . Initially, let $k = 1$, and take the distance from the busbar on the M side to the midpoint of the line as initial fault distance.
- Assume that the line is uniform. Based on the positive-sequence components of the fault quantities on the M side and the N side respectively, formulate the line parameter equations, and calculate the two positive-sequence fault component voltages $\Delta \dot{U}_{M1,F}$ and $\Delta \dot{U}_{N1,F}$ corresponding to the assumed fault point.
- Respective comparisons are made between the phase angles and amplitudes of these two positive-sequence fault-component voltages. Subsequently, the difference in phase angles and the difference in amplitudes are calculated.
- Determine whether the two voltages are equal. If they are, output the currently assumed fault distance. If not, let $k = k + 1$ and continue the iteration.

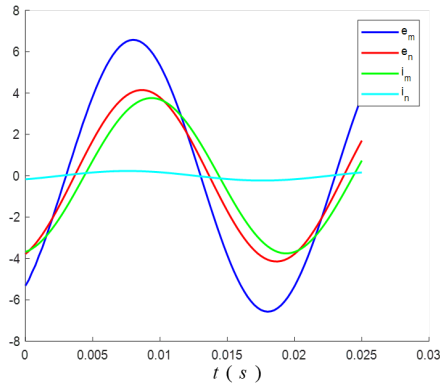


Fig. 3. Three-phase current and voltage diagrams on both sides of the busbar

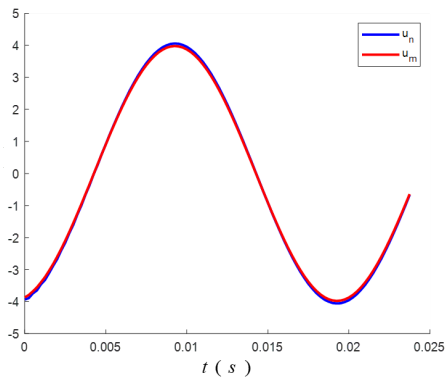


Fig. 4. Waveform diagrams of the two positive-sequence voltages of the fault components at the fault location that satisfy the convergence condition.

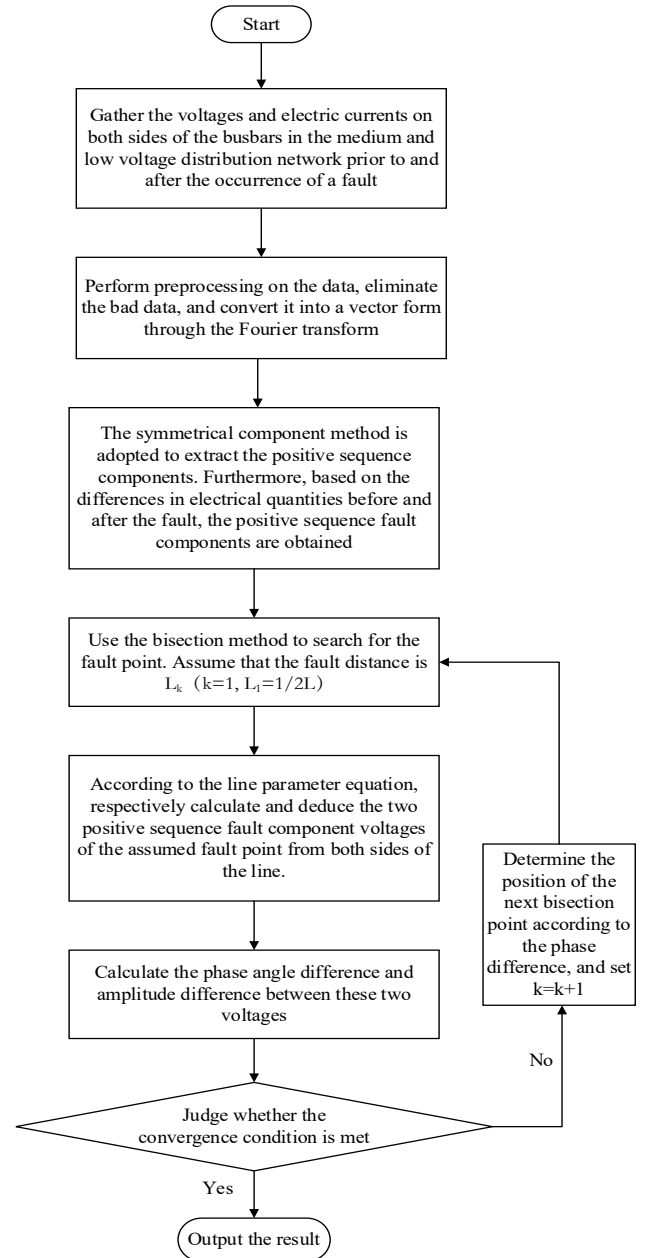


Fig. 5. The flowchart of fault location for distance measurement

IV. SIMULATION ANALYSIS

In the PSCAD/EMTDC simulation software, a distribution network model powered by a double-ended power supply with a voltage level of 10 kV is built, as shown in Fig. 6. The length of the line is 10 km, and a π -type lumped parameter model is adopted. The positive-sequence resistance, inductive reactance, and capacitive reactance per unit length of the line are 0.2 ohm/km, 0.4 ohm/km, and 27.3 M ohm/km respectively, while the zero-sequence resistance, inductive reactance, and capacitive reactance per unit length are 0.2 ohm/km, 1.4 ohm/km, and 41.4 M ohm/km respectively. The sampling frequency is 4 kilohertz, and the sampling window length is 2.5 milliseconds. Fault points are set on the line, and by changing the positions of the fault points and the fault types, the fault location results as shown in TABLE I to Table III are obtained.

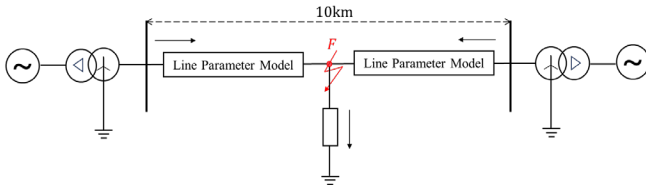


Fig. 6. Schematic diagram of 10kV distribution network

TABLE I. RANGING DATA FOR B-C PHASE-TO-PHASE SHORT CIRCUIT FAULTS

Phase B-C interphase fault		
Set distance of the fault / km	Measured distance / km	Fault location error
1	0.9375	0.625%
3	3.125	1.25%
5	5	0
7	7.03125	0.3125%
9	9.0625	0.625%

TABLE II. RANGING DATA FOR B-C PHASE-TO-PHASE GROUND FAULTS

B-C two-phase ground fault		
Set distance of the fault / km	Measured distance / km	Fault location error
2.5	2.5	0
4.5	4.53125	0.3125%
5.5	5.625	1.25%
6.5	6.5625	0.625%
8.5	8.4375	0.625%

TABLE III. RANGING DATA FOR THREE-PHASE SHORT-CIRCUIT FAULTS

Three-phase short-circuit fault		
Set distance of the fault / km	Measured distance / km	Fault location error
2.1	2.1875	0.875%
4.2	4.0625	1.375%
5.3	5.3125	0.125%
6.0	5.9375	0.0625%
8.4	8.4375	0.375%

In summary, as can be seen from the ranging data shown in TABLE I to Table III, the ranging method proposed in this paper has demonstrated a rather satisfactory ranging effect in medium and low voltage distribution networks. The ranging error does not exceed 2% of the total line length.

V. SUMMARY

Using positive-sequence fault components as its foundation, this paper presents a fault location technology for distribution networks. Grounded on the ranging principle of the double-ended impedance method, it innovatively incorporates positive-

sequence fault components, making it applicable to various types of phase-to-phase faults in medium and low voltage distribution networks. This method is not restricted by the type of fault, nor is it affected by load current, transition resistance, or system potential. It is capable of eliminating the impact of mutual inductance between parallel lines on fault location, ensuring a relatively high ranging accuracy. Thus, it holds significant engineering application value.

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